9th Class • Punjab Board • PECTAA New Syllabus

Chapter 9: Nature of Science

Complete solved exercise — scope and branches of Physics, scientific method, falsifiability, technology and engineering.

Chapter 9 • Complete official exercise • 11 MCQs + 8 short + 10 constructed + 5 comprehensive

What this chapter is about

• **Science** builds testable knowledge of natural phenomena. Physics studies matter, energy, space, time and their interactions.
• **Scientific method** uses observation, questions, hypotheses, predictions, experiments, data analysis and repeated review; it is iterative rather than a rigid recipe.
• **Theories** explain how or why phenomena occur; **laws** summarize observed regularities. Both remain open to refinement when reliable new evidence requires it.
• **Technology and engineering** apply scientific knowledge to tools and designed solutions, often through interdisciplinary collaboration.

Scientific caution: The MCQ wording "ultimate truth" follows the textbook's intended key (law), but modern science treats every explanation as provisional and evidence-dependent. A theory is not a guess, and a law is not a theory that has simply been promoted.

A. Multiple Choice Questions

1. Physics is a branch of:

(a) social science (b) life science
(c) physical science (d) biological science

Answer: (c) physical science — Physics belongs to physical science, which investigates non-living matter, energy and their interactions.

2. Which branch of science plays a vital role in technology and engineering?

(a) Biology (b) Chemistry
(c) Geology (d) Physics

Answer: (d) Physics — Physics supplies fundamental laws of motion, energy, electricity, materials and waves used throughout technology and engineering.

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3. Automobile technology is based on:

(a) acoustics (b) electromagnetism
(c) optics (d) thermodynamics

Answer: (d) thermodynamics — The engine's conversion of fuel energy into mechanical work is governed principally by thermodynamics, though modern vehicles use many branches.

4. A user-friendly software application of a smartphone uses:

(a) laser technology (b) information technology
(c) medical technology (d) electronic technology

Answer: (b) information technology — Software design, data processing and human–computer interfaces are parts of information technology.

5. The working of refrigeration and air conditioning involves:

(a) electromagnetism (b) mechanics
(c) climate science (d) thermodynamics

Answer: (d) thermodynamics — Refrigeration transfers heat using phase changes, compression and thermodynamic cycles.

6. What is the ultimate truth of a scientific method?

(a) Hypothesis (b) Experimentation
(c) Theory (d) Law

Answer: (d) Law — The textbook's intended answer is law: a repeatedly tested general statement of natural behaviour. Scientifically, however, no law is an absolute or unrevisable 'ultimate truth'.

7. "If I do not study for this test, then I will not get a good grade" is an example of:

(a) theory (b) observation
(c) prediction (d) law

Answer: (c) prediction — It predicts a future outcome from a stated condition. It is not a scientific law and has not yet been observed for the future case.

8. Which of the following are methods of investigation?

(a) Observation (b) Experimentation
(c) Research (d) All of these

Answer: (d) All of these — Observation, controlled experiments and wider research are complementary ways of collecting and evaluating evidence.

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9. A hypothesis:

(a) may or may not be testable (b) is supported by evidence
(c) is a possible answer to a question (d) all of these

Answer: (d) all of these — The intended answer is (d): it begins as a possible answer and can gain evidential support. For use as a scientific hypothesis, however, it must be testable and falsifiable; an untestable idea is not thereby proven wrong.

10. A graph of organized data is an example of:

(a) collecting data (b) forming a hypothesis
(c) asking a question (d) analyzing data

Answer: (d) analyzing data — Graphing reveals trends, relationships and anomalies, so it is a method of analysing data already collected.

11. The colour of a door is brown. It is an example of:

(a) observation (b) hypothesis
(c) prediction (d) law

Answer: (a) observation — This is a direct qualitative observation, with no proposed explanation or future prediction.

B. Short Answer Questions

9.1. State in your own words what science is. Write its two main groups.

Science is an organized, evidence-based way of building knowledge about natural phenomena by asking questions, making observations and testing explanations. The textbook's two main groups are **biological sciences**, which study living things, and **physical sciences**, which study non-living matter and energy.

9.2. What is Physics all about? Name some of its branches.

Physics studies matter, energy, space, time and their mutual interactions, from subatomic particles to galaxies. Its branches include mechanics, heat and thermodynamics, acoustics, optics, electromagnetism, quantum mechanics, relativity, nuclear physics, particle physics and astronomy.

9.3. What is meant by interdisciplinary fields? Give a few examples.

Interdisciplinary fields combine concepts, methods and evidence from two or more disciplines to investigate problems that one field alone cannot fully address. Examples include **biophysics**, **astrophysics**, **geophysics**, **climate physics** and **computational physics**.

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9.4. List the main steps of scientific method.

Observe and ask a question; gather background information; form a testable hypothesis; make a prediction; plan and perform an experiment; collect and analyse data; accept, reject or revise the hypothesis; develop broader theories or laws after repeated testing; and communicate results for review and replication. Real investigations revisit steps as needed.

9.5. What is a hypothesis? Give an example.

A hypothesis is a tentative, testable explanation or possible answer to a scientific question, often expressed as an if–then prediction. Example: “If all other conditions are kept the same, increasing a pendulum’s length will increase its time period.” An experiment can support or contradict it.

9.6. Distinguish between a theory and a law of Physics.

Theory	Law
A well-tested explanatory framework describing how or why phenomena occur.	A concise statement, often mathematical, describing a consistent natural relationship.
Examples: kinetic theory and theory of relativity.	Examples: conservation of energy and Newton’s laws within their domains.
Can include models, mechanisms and many related observations.	Summarizes what happens under specified conditions, not necessarily why.
Can be refined or replaced by a broader explanation.	Can acquire revised limits or a more general form with new evidence.

9.7. What is the basis of laser technology?

Laser technology is based on **atomic physics**, especially stimulated emission. An excited atom is triggered to emit a photon matching the stimulating photon; optical amplification produces coherent, nearly monochromatic and highly directional light used in communication, medicine, industry and research.

9.8. What is the falsifiability concept? How is it important?

A claim is falsifiable if an observation or experiment could, at least in principle, show it to be false. Falsifiability makes a claim scientifically testable, exposes it to genuine evidence and helps distinguish science from statements that are adjusted to agree with every possible outcome.

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C. Constructed Response Questions

9.1. Is a theory of science an ultimate truth? Describe briefly.

No. A scientific theory is the best current explanation supported by a substantial body of evidence, not an absolute final truth. It remains open to further testing and may be refined, restricted to a domain or replaced if reliable new observations require a better account.

9.2. Do you think existing laws of nature may need a change in future? Describe briefly.

Yes, their mathematical form or range of applicability may need revision when new reliable evidence disagrees with them. Newtonian mechanics remains extremely useful at ordinary speeds and scales, while relativity provides a more general account at high speed and strong gravity. New laws often preserve older laws as limiting approximations rather than making all previous knowledge useless.

9.3. Describe three jobs that need the use of scientific knowledge.

1. Medical doctor or radiographer: Uses biology, physics and chemistry for diagnosis, radiation safety, imaging and treatment.

2. Civil engineer: Applies mechanics, materials science and fluid pressure to design safe bridges, buildings and water systems.

3. Meteorologist: Uses thermodynamics, fluid motion, radiation and computer models to analyse weather and issue forecasts.

Many other careers—electrician, pilot, pharmacist, agricultural scientist and software engineer—also rely on scientific evidence and quantitative reasoning.

9.4. Describe when a theory is rejected or needs modification.

A theory needs modification when repeated, reliable observations contradict its predictions, when it fails outside a demonstrated range, or when a broader theory explains the same evidence plus new phenomena. Scientists first check equipment, methods and analysis. It is rejected when its central claims fail and repairing them is no longer reasonable or predictive.

9.5. Comment on: “A theory capable of being proved right but not being proved wrong is not a scientific theory.”

The central idea is correct: a scientific theory must take the risk of disagreement with evidence. If every imaginable result is interpreted as support and no possible test could contradict it, the theory is unfalsifiable and cannot be investigated scientifically. Evidence can strongly support a theory without proving it absolutely, while a valid counterexample can demand revision.

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9.6. What has been the general reaction to new ideas about established truths?

New ideas often meet scepticism, debate and resistance because they challenge familiar beliefs, authority or existing practice. Healthy scientific scepticism demands evidence and replication, but unreasonable attachment can delay acceptance. Ideas such as heliocentrism and continental drift gained acceptance as diverse evidence and predictive power accumulated.

9.7. If a hypothesis is not testable, is the hypothesis wrong? Explain.

Not necessarily. “Untestable” means no available observation or experiment can support or contradict it, so it is not a usable **scientific** hypothesis at that stage. It may be true, false or become testable later with better technology or a clearer formulation, but science cannot decide merely by assertion.

9.8. Explain how a small amount of data cannot prove that a prediction is always correct but can prove it is not always correct.

A universal claim covers unlimited cases, places and times. A few confirming examples may be unrepresentative and cannot check every case, so they do not prove “always”. However, one carefully verified case that violates the prediction is a counterexample and disproves the universal claim as written, requiring rejection or limitation.

9.9. What is the relationship between an experiment and a hypothesis?

A hypothesis supplies a testable prediction, and an experiment is designed to compare that prediction with controlled observations. The investigator changes the independent variable, measures the dependent variable and keeps other relevant factors controlled. Results support, contradict or refine the hypothesis; they do not provide absolute proof.

9.10. Describe why solving complex problems needs interdisciplinary research and collaboration.

Complex problems span several mechanisms and scales. Climate action needs atmospheric physics, chemistry, ecology, engineering, economics and social behaviour; medical imaging needs physics, computing, biology and clinical expertise. Collaboration combines specialized knowledge, instruments and data, checks blind spots and turns explanations into practical, ethical solutions that no single discipline could deliver alone.

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D. Comprehensive Questions

9.1. What are the main branches of Physics? State them briefly.

Branch	Main concern
Mechanics	Motion, forces, energy and gravitation; classical mechanics uses Newton’s laws.
Heat and thermodynamics	Thermal energy, temperature, heat transfer and conversion between heat and work.
Acoustics	Production, propagation and reception of audible sound.
Optics	Behaviour of visible light, including reflection, refraction and optical instruments.
Electromagnetism	Electric and magnetic phenomena and their relationship, including electromagnetic waves.
Quantum mechanics	Behaviour of matter and energy at atomic and subatomic scales.
Relativistic mechanics	Space, time, motion and gravity at high speeds and strong gravitational fields.
Nuclear physics	Properties and reactions of atomic nuclei.
Particle physics	Elementary particles and the fundamental interactions between them.
Astronomy	Celestial objects such as planets, stars and galaxies and their distribution.

9.2. What is meant by interdisciplinary fields of Physics? Give three examples.

Interdisciplinary physics integrates physical principles with another subject’s questions, data and methods. It allows physicists to address systems that cannot be understood through a single traditional branch.

1. Biophysics: Applies mechanics, electricity and molecular physics to living systems, such as nerve signals, blood flow, protein structure and medical imaging.

2. Astrophysics: Uses gravity, nuclear physics, radiation and quantum mechanics to explain stars, galaxies and the universe.

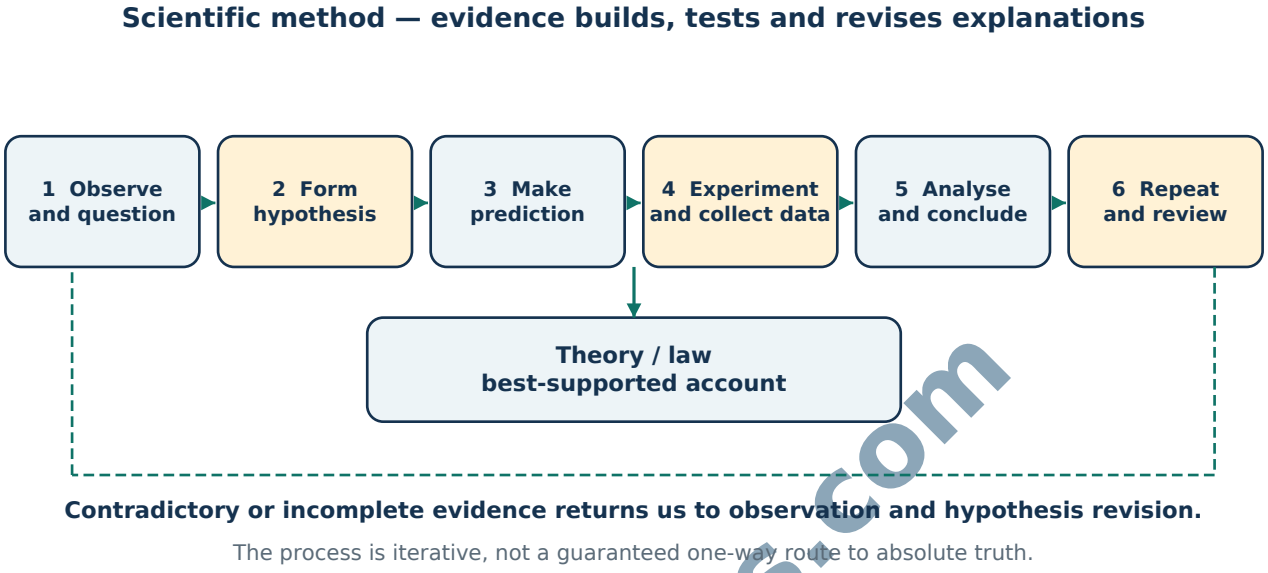
3. Geophysics: Applies seismology, gravity, magnetism and fluid motion to Earth’s interior, earthquakes and natural resources.

Other examples include climate physics and computational physics, which combine atmospheric processes and numerical modelling respectively.

9.3. What is scientific method? Describe its main steps with examples.

Scientific method is a systematic, evidence-based approach for investigating natural phenomena and testing explanations. It combines creativity with controlled comparison, reproducibility and willingness to revise.

Figure 1 — Iterative route from observation to a supported explanation



1. Observation and question: A pendulum seems to swing more slowly when its string is longer. Does length affect its period?

2. Hypothesis: With amplitude and mass fixed, increasing length increases period.

3. Prediction: If length doubles, measured period will follow the hypothesized relationship.

4. Experiment: Change only length, keep release conditions similar, repeat timings and record uncertainties.

5. Analysis: Graph period against length or √length and compare with the prediction.

6. Conclusion and review: Support, reject or revise the hypothesis; communicate methods so others can replicate them.

Repeated successful tests can contribute to a broader theory or law. The steps may loop back whenever an anomaly or new question appears.

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9.4. Differentiate science, technology and engineering with examples.

Figure 2 — Distinct purposes connected in a two-way relationship

Science, technology and engineering work together



Engineering also weighs safety, cost, reliability, materials and human needs.

New tools enable new science; scientific discoveries enable new technologies.

Example chain: atomic physics → laser technology → designed medical equipment.

Field	Purpose	Example
Science	Develops and tests knowledge of natural phenomena.	Atomic physics explains energy levels and stimulated emission.
Technology	Develops methods, techniques and tools using knowledge.	Laser technology produces coherent light for communication and surgery.
Engineering	Designs, builds and improves solutions under practical constraints.	Engineers design a laser surgical system that is safe, reliable and affordable.

Science can advance because better instruments become available, while engineering depends on scientific understanding. Their goals differ, but collaboration is continuous.

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9.5. What is the scope of Physics in everyday life? Give some examples.

Everyday area	Physics involved
Transport	Engine thermodynamics, friction, braking, aerodynamics and vehicle stability.
Communication	Electromagnetic waves, optics, semiconductors and signal processing in phones and fibre links.
Home electricity	Current, resistance, power, magnets in motors and safety devices.
Medicine	X-rays, ultrasound, lasers, nuclear imaging and biomechanics.
Buildings and water	Forces, equilibrium, materials, pressure and energy-efficient insulation.
Weather and environment	Heat transfer, fluid flow, radiation and satellite remote sensing.
Sports and music	Projectile motion, momentum, levers, vibration and acoustics.

Physics therefore supports both understanding and responsible design, from choosing safe tyres to evaluating energy use and technological risks.

E. More Important Questions (from the text)

1. Why are reproducibility and communication important in science?

Reproducibility means that other investigators can follow reported methods and obtain consistent results within understood uncertainties. It exposes accidental errors, biased selection and equipment faults and makes evidence more trustworthy.

Communication through reports and peer review allows methods and reasoning to be questioned, compared and reused. A result that cannot be independently checked has limited scientific value, even if it appears striking.

2. Does scientific knowledge guarantee that a technology will be used responsibly?

No. Science can explain likely effects and engineering can improve safety, but choices also involve economics, law, culture and ethics. Nuclear reactions can support power generation and medicine, while misuse can cause harm.

Responsible innovation therefore requires risk assessment, transparent evidence, regulation, public discussion and consideration of long-term environmental and social effects alongside technical performance.

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Points to Remember

Science groups: biological sciences study life; physical sciences study non-living nature.

Physics: matter, energy, space, time and their interactions.

Hypothesis: tentative, testable explanation or prediction.

Experiment: controlled test that compares predictions with evidence.

Theory: well-supported explanation of how or why; not a mere guess.

Law: concise regularity or relationship, often mathematical; not absolute truth.

Falsifiability: a scientific claim must risk contradiction by possible evidence.

Counterexample: one verified violation defeats a universal claim; a few confirmations do not prove it.

Interdisciplinary work: combines expertise for problems beyond one field.

Science–technology–engineering: understand nature, create methods and design practical solutions.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	d	3	d	4	b	5	d
6	d	7	c	8	d	9	d	10	d

How to attempt this chapter in the paper

• Separate observation from inference: “the door is brown” observes; “it was painted recently” infers.
• Do not call a statement scientific merely because it sounds technical; ask what evidence could refute it.
• Keep independent, dependent and controlled variables distinct when designing an experiment.
• Treat an anomalous result carefully: repeat and check the method before overturning a theory.
• For “always” claims, look for counterexamples and sampling limits rather than relying only on confirmations.
• When revising a model, state its useful range; an approximation can remain valuable after a broader theory appears.